

lated. For that you need to refer sieve supplier's datasheet to know sieve quantity required to produce, say, one

litre per minute (1 LPM) oxygen at specific concentration limit.

For example, 700 grams of sieves

are required for 1 LPM oxygen at 94% concentration. For oxygen concentrator with 5 LPM flow, you need $700 \times 5 = 3,500\text{gm}$ in one canister. So, for two canisters, a total of $3,500 \times 2 = 7,000\text{gm}$ sieves are needed.

Software

Arduino program *oxygen_concentrator_diy.ino* was used for the prototype. It must be uploaded to Arduino Uno. For this, open source software Arduino IDE is required. In the program, Arduino digital pins 2, 3, 4, 5, 6, and 7 are defined as outputs.

Main functions of the Arduino code are:

PinMode(). Configures the specified pin to behave either as an input or an output. See the description of digital pins for functionality of the pins. They can also be used as PWM pins.

DigitalWrite(). If the pin has been configured as an output with `pinMode()`, its voltage will be set to the corresponding value of 5V (or 3.3V on 3.3V boards) for high, and 0V (ground) for low.

Serial. Used for communication between the Arduino board and a computer or other devices. All Arduino boards have at least one serial port (also known as a UART or USART), and some have several.

Begin(). Sets the data rate in bits per second (baud) for serial data transmission. For communicating with the computer, use one of these rates: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, or 115200. You can, however, specify other rates, for example, to communicate over pins 0 and 1 with a component that requires a particular baud rate.

Println(). Prints data to the serial port as human-readable ASCII text followed by a carriage return character.

Delay(). It is used to pause the program for the amount of time (in milliseconds) specified as parameter.

As shown in Fig. 1, six 12V DC SPDT relays are used to control six valves (V1 through V6 via their relay contacts) connected across their contacts. Six fly-wheel diodes (D1 through D6) are used across relays to prevent the damage of transistors from back EMF.

The Arduino is programmed to allow compressed air to enter from one of the canisters at a time. Some

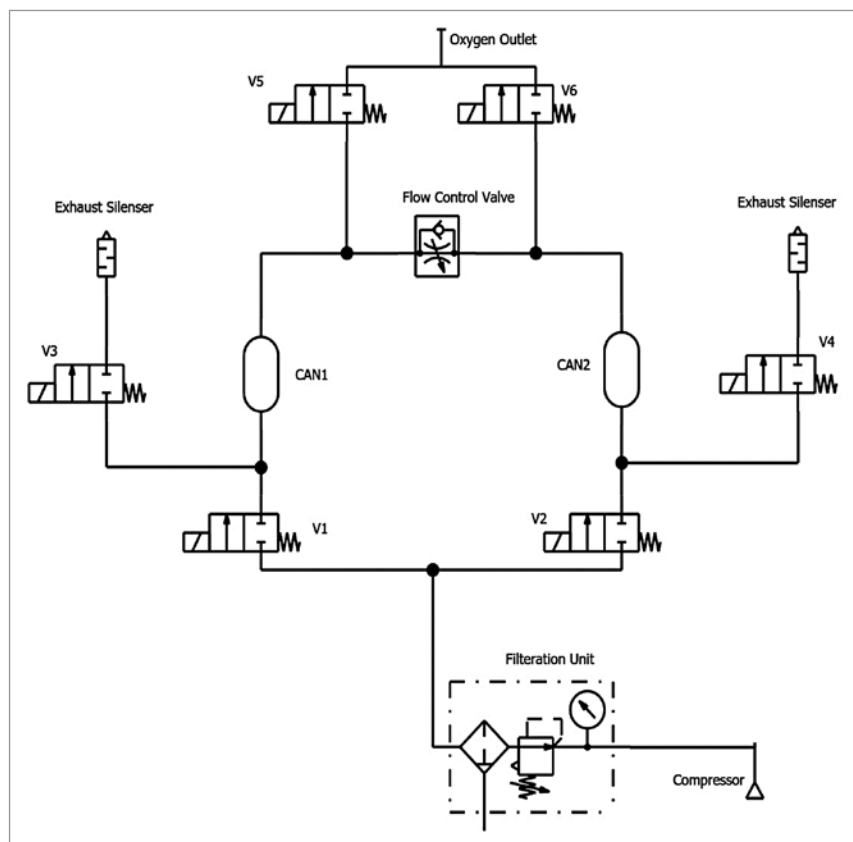


Fig. 4: Pneumatic arrangement

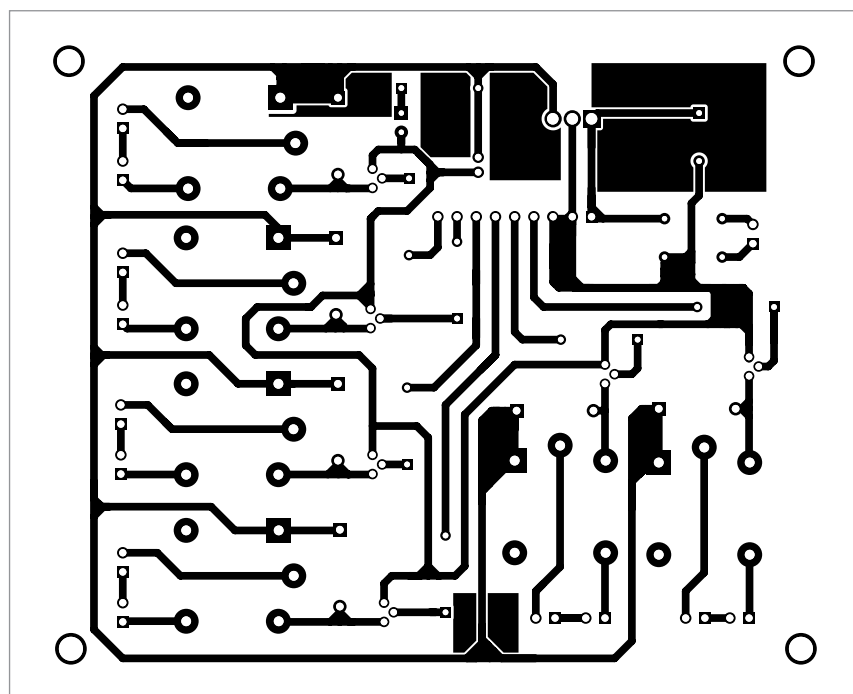


Fig. 5: Actual-size PCB layout for oxygen concentrator controller